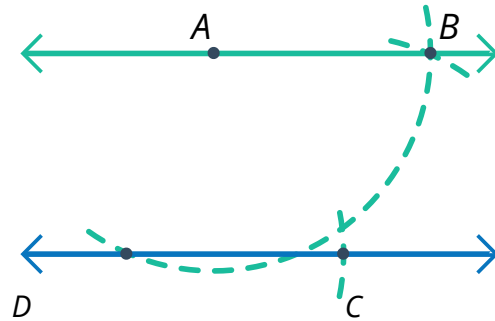


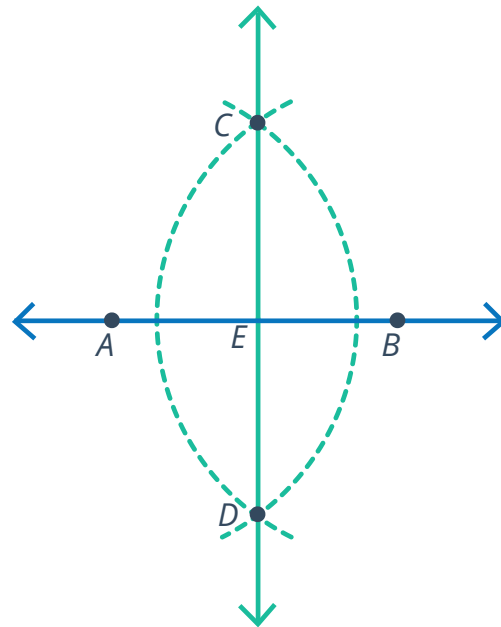
## Justifying constructions using the properties of quadrilaterals

1 Consider the following diagrams. Justify the steps of the construction in a two-column proof.

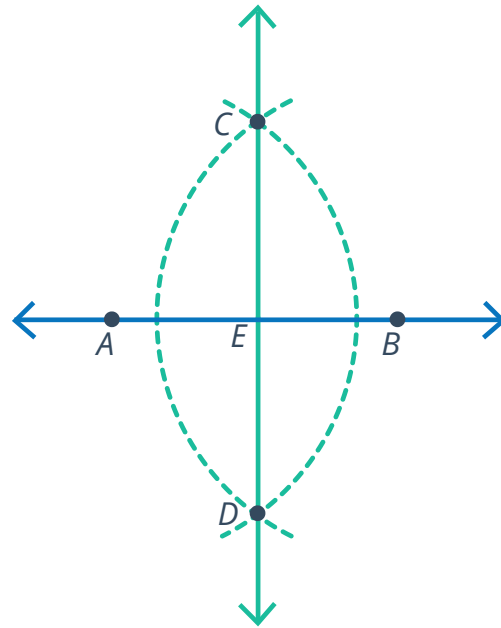
- a The line  $\overleftrightarrow{AB}$  is constructed parallel to the given line  $\overleftrightarrow{DC}$ .



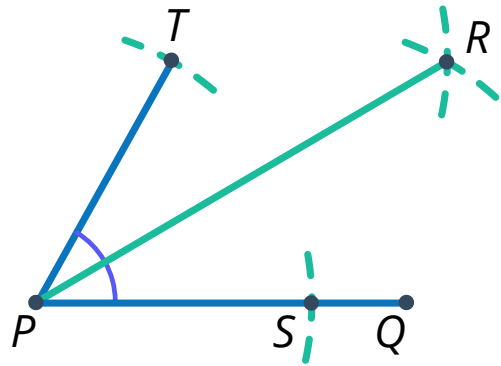
- b The line  $\overleftrightarrow{CD}$  is constructed so that  $\overleftrightarrow{AB} \perp \overleftrightarrow{CD}$ .



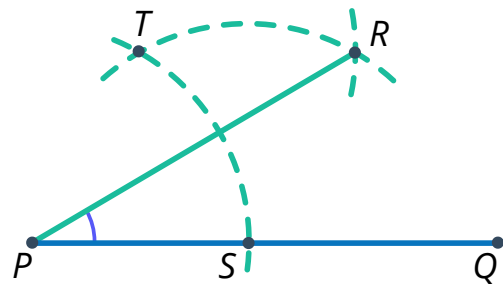
- c The line  $\overleftrightarrow{CD}$  is constructed so that  $\overline{AE} \cong \overline{BE}$ .



- d  $\overline{PR}$  is constructed so that  $\angle TPR \cong \angle RPQ$ .



- e  $\overline{PR}$  is constructed from the given line segment  $\overline{PQ}$  such that  $m\angle RPS = 30^\circ$ .



- 2 The inscribed square  $ABCD$  is constructed from a given circle with a center at the point  $O$ . Justify the steps of the construction in a paragraph proof.

